

Assignment Solutions- Probability, CLT and Distributions

1. Two fair dice rolled.

Total possible outcomes: $6 \times 6 = 36$

Interview

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Favorable cases: dice 1 shows one, dice 2 shows more than 3 = 3

Dice 1 shows two, dice 2 shows more than 2 = 4

Dice 1 shows three, dice 2 shows more than 1 = 5

Dice 1 shows 4, dice 2 shows anything = 6

Dice 1 shows 5, dice 2 shows anything = 6

Dice 1 shows 6, dice 2 shows anything = 6

Total favorable cases = $3+4+5+6+6+6 = 30$

2. Probability = $p(\text{first ball is blue}) \times p(\text{second ball is orange} | \text{first ball is blue}) \times p(\text{third ball is red} | \text{first ball is blue and second ball is orange})$

$$= \frac{5}{12} \times \frac{3}{11} \times \frac{4}{10}$$

3. $P(\text{ball drawn is from the third bag} | \text{ball drawn is red}) = \frac{P(\text{ball drawn is from the third bag} \cap \text{ball is red})}{P(\text{ball drawn is red})}$

$P(\text{ball drawn is from the third bag} \cap \text{ball is red}) = \frac{1}{3} \cdot \frac{4}{10} = \frac{2}{15}$

$P(\text{ball drawn is red}) = \frac{1}{3} \cdot \frac{3}{10} + \frac{1}{3} \cdot \frac{8}{10} + \frac{1}{3} \cdot \frac{4}{10} = \frac{1}{2}$

$P(\text{ball drawn is from the third bag} | \text{ball drawn is red}) = \frac{(\frac{2}{15})}{(\frac{1}{2})} = \frac{4}{15}$

4. Number of days in a leap year = 366 = 52 weeks + 2 days.

Till 364 days, everyday must have come exactly 52 times.

365th day	366th day	1st day (next year)	2nd day	3rd day	4th day	5th day
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So for Sunday to occur only 52 times, it shouldn't lie in the first two boxes. So probability = 5/7

5. Here we have three possible case:

Door 1	Door 2	Door 3	If switch the door	If door is not switched
Goat	Car	Goat	Win	Lose
Goat	Goat	Car	Win	Lose

Car	Goat	Goat	Lose	Win
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As one can see, if you switch the door you are more likely to win the same (i.e. with 2/3rd probability)

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6. For a normally distributed data with mean 100 and sd 15, z score for $x > 130$ is $130-100/15 = 2$
As we know $P(-1.96 < z < 1.96) = 95\%$,
Hence, $P(z > 2) + P(z < -2) = 5\%$, i.e. $P(z > 2) = 2.5 \%$
7. Expected value of a random variable is the weighted average of all possible values of the variable. This is also called the mean of a random variable. Hence, the answer is $E(X)$.
8. The probability $P(0.5 < x < 5)$ can be splitted into three parts, $P(0.5 < x < 1) + P(1 < x < 4) + P(4 < x < 5) = 0.2 + 0.1 + 0 = 0.3$
9. Standard deviation is a measure of distance hence it is always positive.
10. $P(x > (\mu + \sigma)) = (1 - P(z=1))/2 = (1-0.682)/2 = 0.318/2 = 0.159$. Hence, on left of $(\mu + \sigma)$, probability is $= 1-0.159 = 0.841$
11. The area under a CDF represented the universe of outcomes and is the sum of the total probability equal to 1
12. Follows from the theorem of independent variables and expected values
13. Two random variables X and Y are said to be jointly normal if they can be expressed in the form
 $X = aU + bV$,
 $Y = cU + dV$
a) $(aU + bV) + (cU + dV) = (a+c)U + (b+d)V$
is jointly normal with $(aU + bV) - (cU + dV) = (a-c)U + (b-d)V$
as both can be expressed as linear combination
b) Use example 14
c) If two random variables X and Y are jointly normal and are uncorrelated, then they are independent

14. $\text{Cov}(X+Y, X-Y) = \text{Cov}(X+Y, X) + \text{Cov}(X+Y, -Y)$
 $= \text{Cov}(X, X) + \text{Cov}(Y, X) + \text{Cov}(X, -Y) + \text{Cov}(Y, -Y)$
 $= 1 + \text{Cov}(X, Y) - \text{Cov}(X, Y) - 1$
 $= 0$
15. There is no assumption about where (specific floor) and when (together or separately) people get on the elevator.

Probability of a person getting off at a specific floor (out of 10) = $1/10$

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Probability a person not getting off at a specific floor = $1 - 1/10 = 9/10$

Probability none of 8 get off at a specific floor = $(9/10)^8$ = Probability of not stopping at a specific floor (assuming all people are independent)

Probability of stopping at a specific floor = $1 - (9/10)^8$

Expected number of stoppings for 10 floors = $E(x) = 10 * (1 - (9/10)^8)$

16. Median is better when data has outliers or extreme values. For example, {5, 6, 12, 13, 2,

19, 20, 99999}

17. The number of people showing up during a four hour period is a Poisson random variable X with mean $4(2.5) = 10$.

$$P(X = 0) = e^{-10} (10^0)/0!$$

$$P(X = 1) = e^{-10} (10^1)/1!$$

$$P(X = 2) = e^{-10} (10^2)/2!$$

$$P(X = 3) = e^{-10} (10^3)/3!$$

Hence, probability is $e^{-10} [1 + 10 + 50 + 1000/6] = \sim 0.0103$.

18. For exponential distribution, $P(x < t) = 1 - e^{-t}$. Hence, in this case, $P(X > t) = e^{-t} = e^{-5/3}$

19. By the definition of it, Poisson is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event

20. A binomial trial is defined on the basis of the probability of outcome and hence it cannot change from one trial to another. p & q are fixed throughout the experiment.

21. Apply Bayes' Theorem to solve the problem,

$$P(A | B) = P(B | A) \times P(A) / P(B)$$

A = Probability of the first toss is head = $\frac{1}{2}$

B = Probability of 4 heads in 10 tosses = ${}^{10}C_4 * (1/2)^4 * (1/2)^6 = 210 * (1/2)^{10}$

$P(B|A)$ = Probability of 4 heads in 10 tosses given that first is head =

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$$3 * (1/2)^9 = 84 * (1/2)^9$$

$${}^9C_3 * (1/2)^9 = (\text{Probability of getting 3 heads in remaining 9 tosses})$$

$${}^9C_3 * (1/2)^9 = 210 * (1/2)^9$$

$$P(A|B) = 84 * (1/2)^9 / 210 = 2/5$$

22. Let x be the probability of a car passing the intersection in a 5-minute window.

Probability of a car passing in a 20 minute window = $1 - (\text{probability of no car passing in a 20 minute window})$

Probability of a car passing in a 20 minute window = $1 - (1 - \text{probability of a car passing in a 5 minute window})^4$

$$0.9 = 1 - (1 - x)^4$$

$$(1 - x)^4 = 0.1$$

$$1 - x = 10^{(-0.25)}$$

$$x = 1 - 10^{(-0.25)} = \mathbf{0.4377}$$

23. Random variable created by addition of 2 random variables is again a **normal random variable**.

For the probability of the product of $X(0,1)$ and $Y(0,1)$ being less than 0.5, a quick way to go about it is to visualise a 2 dimensional plane. All the points (x,y) within the square $[0, 1] \times [0, 1]$ fall in the candidate space. The case when $xy = 0.5$ makes a curve $y = 0.5/x$. The area under the curve would represent the cases for which $xy \leq 0.5$. Since the area for the square is 1, that area is the sought probability.

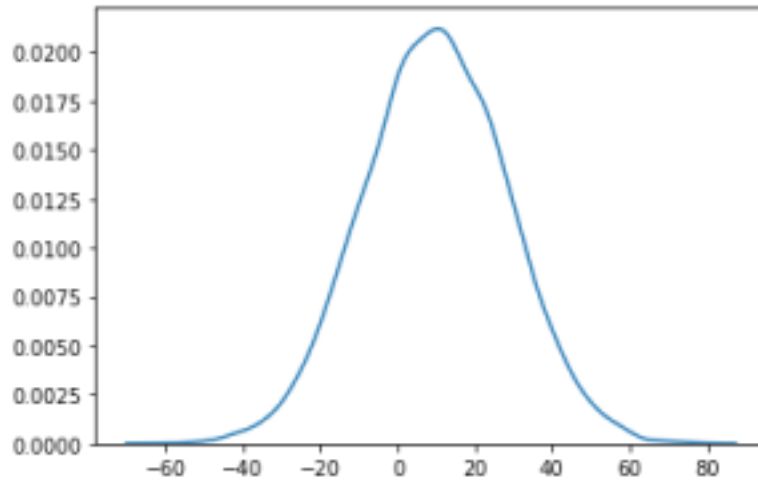
The curve intersects the square at $[0.5, 1]$ and $[1, 0.5]$.

The area under the curve would be the sum of the 2 quadrants ($1/4$ each) along with the integral of $y = 0.5/x$ under the range 0.5 to 1 yielding

$$= 1/2 + \int_{0.5}^1 (0.5/x) dx = 0.85 (\text{approx})$$

24. For John, sample space consists of all possible free seats which have only two elements. Either the seat intended for the Mark is free or the last passenger's seat is available. As the chance for both events are equal and they sum up to 1, hence required probability is $\frac{1}{2}$

25. Z is a normal variable with the distribution depicted below



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The above plot depicts a bell-shaped curve and it is in conjunction with our hypothesis that $Z = X + Y$ would represent a normal distribution

26. Let A denote the number of heads that you got and B the number of heads your friend has got. Then

$$P(A=0, B=3) = P(A=3, B=0) = 1/8$$

$$P(A=1, B=2) = P(A=2, B=1) = 3/8$$

Hence,

$$P(A=0, B=0) = P(A=3, B=3) = 1/8 * 1/8 = 1/64$$

$$P(A=1, B=1) = P(A=2, B=2) = 3/8 * 3/8 = 9/64$$

When you add up these four, you will get the required probability which is $5/16$.

Appendix:

Older version of 23: You have two uniformly distributed random variables x and y. What distribution does the sum of these two random numbers follow? What is the probability that their product is less than 0.5?